



The Story of Your Life: Large Language Models and Personal Memory

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Abstract

The present paper explores how large language models (LLMs) can function as personal memory systems. Unlike conventional lifelogging technologies, which passively record data, LLMs can actively participate in memory processes through dialog, interpretation, and narrative generation. LLMs, we suggest, mark a new chapter in the history of memory technology, one characterized by the active solicitation of mnemonically-relevant information and the co-production of narrative artifacts. Courtesy of their conversational capabilities, LLMs are poised to influence memory in the manner of a human social companion. This, we suggest, blurs the distinction between technologically- and socially-situated approaches to the shaping of human memory.

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1 Introduction

From the earliest knots in strings and marks on clay tablets to the invention of writing and the printing press, human beings have long relied on external resources to shape and augment memory (Clowes 2017; Heersmink [in press](#)). These material supports have historically functioned as scaffolds for individual and collective recall, enabling the preservation and transmission of knowledge across generations. More recent developments—including photographs, phonographs, digital computers, smartphones, search engines, and cloud-based storage—have significantly expanded the fidelity, flexibility, and immediacy of memory storage, providing new opportunities for mnemonic augmentation (Clowes 2013, 2015). Lifelogging technologies have emerged as a particularly important focus area for memory-related research (Dingler et al. 2021; Harvey et al. 2016). Such devices are designed to record and archive the fine-grained details of daily life, ranging from conversations and movements to visual and auditory experiences. As noted by Kalnikaite et al. (2010), these technologies offer a compelling vision of “prosthetic memory”: a comprehensive, searchable archive of lived experience, capable of supplementing or even supplanting biological recall. They also provide important opportunities for therapeutic interventions designed to tackle memory impairments caused by injury or disease (e.g., Dubourg et al. 2016).

In this paper, we suggest that large language models (LLMs) mark a new chapter in the history of technologically-augmented memory. Whereas lifelogging systems primarily function as passive repositories that capture and store information about lived experience, LLMs offer a more active and participatory mode of memory support. Courtesy of their dialogic capabilities, LLMs can act as conversational partners, engaging users in dynamic exchanges that contribute to the formation, retrieval, and enrichment of mnemonic records. In addition, the generative capabilities of LLMs support the co-creation of personalized narratives, helping to weave the elements of lived experience into an evolving life story.

Our aim in the present paper is to explore the status of LLMs as memory technologies. In Sect. 2, we explore some of the innovations that enable LLMs to function as personal memory systems. In particular, we focus on the innovations that are driving the development of real-world memory systems, such as MemoryBank (Zhong et al. 2024), Memoro (Zulfikar et al. 2024), and OS-1 (Xu et al. 2024). Section 3 explores the ways that LLMs can contribute to memory processes, focusing on their role in encoding and retrieval processes. Section 4 briefly considers the theoretical significance of these developments, focusing on the implications for extended and embedded approaches to human memory. Section 5 turns to the question of a ‘good’ memory, discussing LLM hallucinations and their relevance to personal memory. We conclude (in Sect. 6) by highlighting some of the ways that LLMs mark a new direction for philosophical and empirical research into technologically-augmented memory.

2 Large Language Models and Personal Memory

LLMs are AI systems that specialize in language-related tasks, such as text generation, translation, summarization, and question-answering. As with many contemporary AI systems, LLMs are driven by deep (multi-layer) neural networks that are trained on large datasets. Notable examples of LLMs include OpenAI's GPT-5, Meta's LLaMA, and Google's Gemini system. These models exhibit exceptional levels of performance across a wide range of natural language processing tasks. From summarization and translation to creative writing and dialogue systems, LLMs are reshaping how machines understand and generate human-like language, often achieving results that rival or surpass human performance on standard benchmarks (e.g., Shultz et al. 2025).

Recent work has explored the intersection between LLMs and human memory from at least two perspectives. One strand focuses on the relationship between human bio-memory and memory-related mechanisms in LLM architectures. Dong et al. (in press), for example, review attempts to treat Memory-Augmented LLMs (MA-LLMs) as computational models of human episodic memory, evaluating how well such systems capture properties such as event segmentation, selective encoding, temporal contiguity, and dynamic updating. A second strand of research focuses on the use of LLMs to support human memory, either through the development of mnemonic aids (e.g., Maniar et al. 2025) or (more commonly) through the development of personalized conversational assistants (e.g., Xu et al. 2024). In principle, there is no reason why these two strands of research cannot overlap. Indeed, some attempts to support human memory explicitly rely on the integration of human-like memory mechanisms. The MemoryBank system, for example, incorporates a memory updating mechanism inspired by the Ebbinghaus Forgetting Curve (Zhong et al. 2024). Notwithstanding these overlaps, our focus in the present paper relates to the second strand of research: the use of LLMs to support personal memory.

To clarify our use of the term, we understand personal memory to be a form of memory that concerns the encoding, organization, and retrieval of personal data—information about one's own experiences, relationships, activities, and life events. It encompasses emotionally rich, self-referential episodes; semantic information about one's past; and the narrative structures that integrate this information into a coherent sense of identity and temporal continuity. In this broad sense, biological memory already qualifies as a form of personal memory—it is, after all, a form of memory that centers on an individual's personal past.

The primary challenge for LLM-based personal memory systems is to integrate personal data into an LLM's responses—to somehow tailor the model's activity to reflect previously encoded information. One way to achieve this is to rely on standard machine-learning techniques, such as fine-tuning. In practice, however, this approach is fraught with difficulties. The parameters of most LLMs are frozen once their initial training is complete, meaning that their internal weights cannot be modified at inference time. And even when fine-tuning is technically feasible, it is often imprac-

tical due to the risk of catastrophic forgetting¹ and the overheads associated with data preparation. Additional problems relate to issues of transparency, control, and mnemonic auditing: once information has been absorbed into a model's parameter space, it cannot be selectively inspected, edited, or removed. These limitations make conventional fine-tuning techniques an awkward fit for systems that must remain flexible, transparent, and responsive to ongoing changes in an individual's personal history.

How, then, can LLMs act as a technological support for personal memory? As we shall see, the answer comes in the form of a distinction between what is called parametric (or internal) and non-parametric (or external) memory. In essence, LLMs are able to interact with external bodies of information (e.g., a repository of person-specific information), factoring this information into their generative routines. The distinction between parametric and non-parametric memory serves as the basis for a further distinction between what we will call parametric and non-parametric learning. While parametric learning involves adjustments to an LLM's internal parameters (i.e., weight matrices), non-parametric learning involves adjustments to the content of an external memory store. Both types of adjustment are apt to change an LLM's responses, but they do so via different means.

Two additional innovations are relevant to our understanding of LLMs as personal memory systems. The first is a capacity for multimodal data processing. This is important when it comes to dealing with the sort of data delivered by conventional lifelogging devices. Wearable cameras, for example, deliver information in the form of images, while health-tracking devices (e.g., Fitbits) collect information about physiological, behavioral, and environmental variables. Typically, this information needs to be transformed into something that is more amenable to linguistic processing (e.g., a textual description of an image). The advent of multimodal LLMs (e.g., GPT-4V) helps to meet this challenge, enabling disparate lifelogging data to be converted to a common textual format.

Another important innovation relates to tool use. This refers to the capacity of LLMs to interoperate with a wider ecology of computational systems (including other AI systems). This is important because LLMs are mostly intended for language processing, and, more specifically, language generation. Accordingly, they sometimes perform poorly on tasks that require mathematical calculations or logico-deductive reasoning (see Floridi and Chiriatti 2020). An ability to interoperate with external tools or systems helps to address this problem, enabling LLMs to integrate external computational tools into their own generative routines.

We thus have three innovations that are relevant to understanding the status of LLMs as personal memory technologies: multimodal capabilities, external memory and tool use. The remainder of this section offers a brief overview of these innovations.

¹ Catastrophic forgetting is a phenomenon whereby learning causes the model to lose previously acquired knowledge.

2.1 Multimodal Capabilities

While early LLMs were constrained to purely textual inputs and outputs, recent advances have produced a new generation of systems capable of processing and generating data across a range of modalities, including images, audio, video, and sensor streams. These multimodal LLMs extend the interpretative and generative reach of language models, supporting tasks such as image captioning, visual question answering, and cross-modal reasoning.

In talking of multimodal LLMs, we should make a distinction between LLMs that deal with multimodal data via a single architecture (e.g., a transformer architecture) and those that rely on interactions with other generative AI systems, such as variational encoders and diffusion models. The former are what might be called *unified multimodal LLMs*, while the latter are what might be called *modular multimodal LLMs*. One example of a unified multimodal LLM is OpenAI's GPT-4V model. This model integrates visual processing into the LLM's transformer architecture, allowing the model to produce a textual description of the entities depicted in an image. One example of a modular multimodal LLM comes in the form of LLaVA (Liu et al. 2023), which combines an LLM with a specialized machine vision module.

In addition to the distinction between unified and modular multimodal LLMs, there is a further distinction between systems that accept multimodal inputs and those that produce or generate multimodal outputs. The aforementioned GPT-4V model accepts text and images as input and generates text as an output. The appeal to multimodality thus occurs in respect of the model's *inputs*. Other LLMs, however, can produce non-text content (such as images, audio, and videos) as outputs. For example, an LLM might generate an image based on a textual description. In most cases, these systems rely on additional generative AI systems to produce the actual content. OpenAI's DALL-E (version 2), for example, relies on a transformer architecture for text encoding and a diffusion model for image generation.

Multimodal capabilities are an ongoing focus of contemporary research efforts, with newer systems exhibiting increasingly sophisticated abilities. Google's Veo system, for example, is a state-of-the-art video generation model that accepts a combination of image and text prompts (as input) and produces realistic video and audio data (as output).² At first sight, this sort of capability might seem irrelevant to the status of LLMs as personal memory systems. After all, it is natural to think that the main objective for an LLM-based personal memory system would be the generation of linguistic outputs. As we will see, however, the importance of LLMs goes beyond their ability to transform multimodal data into a natural language format. In addition to such capabilities, LLMs can participate in the generation of narrative artifacts with multimedia content, similar to the multimedia biographies that have featured in dementia-related research (see Smith et al. 2009).

² See <https://deepmind.google/models/veo/>.

2.2 External Memory

To function as a personal memory system, an LLM needs some way to store personal data and coordinate its responses with respect to that data. In short, LLMs require a capacity to assimilate new data into their generative routines. As noted above, such forms of assimilation can occur via the adjustment of two sorts of memory: parametric (or internal) memory and non-parametric (or external) memory. Internal memory refers to the information stored within the model's internal parameters (i.e., the weights of a deep neural network, specifically a transformer architecture). This sort of memory is acquired as a result of the model's training regime. If, for example, a model has been trained on a dataset that includes information about the Museum of Modern Art (MoMA), then the model will be in a position to answer questions about MoMA once the training is complete. If, for example, the model is asked about the address of MoMA, the model may respond by saying that "MoMA is on 53rd Street."

The problem with internal memory is that it is both fallible (the model may fail to recall certain information) and difficult to update. The contents of internal memory are encoded in a highly distributed fashion, possibly across billions of parameters, and such distributed forms of encoding make it difficult to update specific items of information. If, for example, the address of MoMA should change, the only option is to retrain the model on the new data. In general, distributed forms of encoding make it difficult (if not impossible) to make targeted updates to specific information items (e.g., those pertaining to MoMA).

Enter the notion of external memory. External memory refers to a data repository (typically a vector database) that lies external to the LLM. When the LLM responds to a user query, it can access this data repository, relying on a retrieval loop to return query-relevant information. The retrieved information can then be used to tailor (or 'condition') the LLM's response.

As a means of making this clearer, suppose that an LLM was trained on data that did not include any information about MoMA. When this LLM is presented with a question such as "Where is MoMA located?", the model may respond by saying that it doesn't know, or it may report an incorrect response. Now suppose that we create a repository of information about MoMA, using information extracted from the museum's website. This repository serves as the LLM's external memory. When the aforementioned query is presented to the LLM, the LLM retrieves information from the repository using the query as a 'retrieval cue'. The execution of this retrieval action then returns information to the LLM, where it is used as context for the LLM's output. The upshot is that the LLM's response is now coordinated with respect to two items of information: 1) the user's original query, and 2) the information retrieved from external memory. Note that the latter type of information (the information retrieved from external memory) could be text that specifies the location of MoMA (e.g., "The main entrance is on 53rd Street, close to Sixth Avenue and just east of our flagship Museum Store"). In response to this enriched context, the LLM is now able to issue the correct response to the user's query: "MoMA is located on 53rd Street." The model is able to produce the correct answer, because (in this case) the answer to the user's query is already contained in the context that drives the model's response. One way to think about this is to regard the external memory as a form of long-term

memory, albeit one that trades in symbolic, linguaform encodings of domain-relevant information. The call to external memory then functions as a retrieval loop that returns information relevant to the current generative context. The combination of the user query and the retrieved information then acts as a sort of ‘working memory’ that drives the LLM’s responses.

This approach to incorporating external information into an LLM’s generative routines is known as Retrieval-Augmented Generation (RAG) (Lewis et al. 2020; Gao et al. 2024). The specific details of this technique need not concern us here.³ What matters is simply the idea that an LLM can access external bodies of information and factor this information into its generative routines. This capability is central to understanding how LLMs can function as personal memory systems. While the parameters of a production-level LLM,⁴ such as ChatGPT, might be fixed, the contents of external memory are not so constrained. As new information is acquired, it can be added to external memory, thereby allowing LLM responses to be conditioned on the newly acquired information. Suppose, for the sake of example, that you are equipped with a wearable camera that periodically takes a picture of your field of view. This image can be posted to a multimodal LLM that generates a textual description of the image, and the textual description can then be added to external memory. If one of the images depicts you engaged in a certain activity (a canoeing activity, for example), then information about this activity will now be stored as a record in external memory. If you subsequently ask the LLM whether you have been engaged in a canoeing activity, the LLM will use your query as a retrieval cue, attempting to retrieve relevant information from its external memory. Assuming the retrieval routine succeeds, information about your canoeing trip will be made available to the LLM, and the LLM will generate its response on that basis.⁵

This, in outline form, is how an LLM can be made to work as a personal memory system without the need for conventional (i.e., parametric) forms of machine learning. Indeed, the learning that occurs in RAG contexts is via the modification of external memory, not the adjustment of the LLM’s internal memory. This is what we earlier referred to as non-parametric learning. A key feature of non-parametric learning is that it allows a single production-level LLM (e.g., ChatGPT) to function as the core of multiple personal memory systems, with person-specific data stores providing the means by which an LLM’s responses are tailored to an individual’s personal history. This is how the majority of experimental LLM-based personal memory systems (e.g., MemoryBank, OS-1, and Memoro) are implemented. It is also the approach adopted by a number of production-level LLMs that rely on a rudimentary form of personal memory to tailor responses to particular users. In the case of ChatGPT, for example, user accounts are associated with a set of “Saved Memories” that

³ See Smart et al. (2025), for an accessible introduction to RAG techniques and their application to the development of ‘philosophically-minded’ LLMs.

⁴ A production-level LLM is a model deployed for real-world use, typically with a fixed set of learned parameters and a stable architecture. Examples include ChatGPT, Gemini, and Grok.

⁵ In practice, retrieval-augmented generation is rarely as straightforward as this idealised description suggests. Even when the retrieval step returns the relevant information, that information must still be integrated into the model’s output. Zhang and Zhang (2025) provide a recent summary of efforts to improve the reliability of RAG techniques (see also Sect. 5).

store information about the user's activities, preferences, interests, and social relationships.⁶ While ChatGPT is not explicitly designed as a personal memory system, the "Saved Memories" feature illustrates how a single production-level LLM can function as the core of multiple personal memory systems, with tailored responses emerging from the content of person-specific data stores rather than from adjustments to the model's internal parameters.

2.3 Tool Use

While RAG represents a powerful technique for extending the capabilities of LLMs, it is increasingly being generalized into a broader class of what are called *augmented LLMs* (Mialon et al. 2023). The distinctive feature of RAG is the inclusion of a retrieval loop that targets a body of domain- or person-specific information, typically in the form of a vector database. In principle, however, there is no reason why the retrieval loop could not be directed at other types of resources. Nor is there any reason to assume that every type of interaction between an LLM and its wider environment is best understood as a form of retrieval action. At a general level, we can characterize RAG as the instantiation of an information processing circuit (or information processing loop) that extends beyond the borders of the LLM. In the case of RAG, the LLM uses (and sometimes generates) content that functions as a retrieval cue. In other cases, however, the LLM might generate content that enables it to invoke an external tool, such as an online calculator or logic reasoner. This is the basic vision behind the notion of augmented LLMs. It is a vision in which the capabilities of LLMs are augmented courtesy of their interaction with external resources or tools. By directing attention to the nature of these interactions, we are able to distinguish different types of augmentation. RAG is, of course, one example, but other forms of augmentation have been the focus of recent research efforts. These include the following:

- **Knowledge-Augmented Generation (KAG).** This form of augmentation closely resembles RAG, except the external memory consists of a graph database containing structured knowledge, typically in the form of (computational) ontologies (see Pan et al. 2024). The specific virtue of ontologies lies in their capacity to support symbolic reasoning and the retrieval of information that respects certain semantic constraints. In this sense, KAG is an ideal candidate for semantic memory applications.
- **Search-Augmented Generation (SAG).** SAG integrates real-time search capabilities into the generative process, typically via an online search engine, such as Google Search. This technique has become something of a standard feature of production LLMs, such as ChatGPT and Gemini. When prompted for news on current affairs, for example, ChatGPT issues a call to a search tool and factors the results into its response. In a personal memory context, this capability can be used to enrich the information obtained from various recording devices. If, for example, a GPS device indicates that an individual is currently in the vicinity of

⁶ See <https://openai.com/index/memory-and-new-controls-for-chatgpt/>.

MoMA, an LLM can perform a search operation to obtain information about the current weather conditions at this location.

- **Tool-Augmented Generation (TAG).** TAG refers to situations where LLMs interact with external computational tools, such as calculators, formal logic reasoners, and online services (Schick et al. 2023). This interaction is achieved courtesy of the LLM's capacity to generate programmatic instructions that respect the target tool's Application Programmatic Interface (API). An LLM might, for example, invoke the services of a specialized reasoning agent to infer information about the temporal relationships between certain events. Such capabilities are apt to be useful when it comes to the process of emplotment, which refers to the temporal ordering of events, activities, and other occurrent entities within a narrative structure (see Heersmink 2022a).
- **Code-Augmented Generation (CAG).** Some LLMs can generate executable code to perform operations that are too complex to be handled by language processing alone (e.g., Gao et al. 2023). Consider a case where an LLM is tasked with computing the total number of steps performed by an individual within a given time period. This requires the LLM to engage in a process that it may not be optimized for (i.e., a task involving numerical reasoning). One possibility is to resort to an external tool, such as a calculator. But another strategy is to generate the (programmatic) code that implements the desired operation. This code can then be executed by a code interpreter and the result returned to the LLM. The total number of steps can then be added to external memory. In some ways, of course, CAG is similar to TAG, in the sense that the code interpreter might be seen as an external tool. On the other hand, CAG introduces us to the idea of LLMs as the *creators* of tools. By generating the code to perform a particular task, the LLM is, in effect, creating an ad hoc computational tool—one that is tailored to the task at hand and (usually) abandoned once the task is complete.

These different forms of augmentation tend to complicate the rather simplistic image of LLMs as the sole technological ingredient of a personal memory system. This is because such systems are likely to involve a multiplicity of technological and informational resources. In addition, there is no reason to assume that personal memory systems will be built around a *single* LLM. In practice, a personal memory system is likely to involve multiple LLMs, with different LLMs being invoked to perform different tasks. A multimodal LLM, for example, might be tasked with the processing of physiological data (see Fang et al. 2024), while another LLM might be tasked with the optimization of retrieval cues (see Ma et al. 2023). Other LLMs may specialize in the generation of narrative artifacts, such as textual descriptions of daily events or significant life periods.

As noted by an anonymous reviewer, the use of augmented generation techniques in combination with multi-LLM solutions complicates the implementation of LLM-based memory systems. In particular, it raises questions about the coordination of LLM activity at both the individual and collective level. At the individual level, a single LLM may need to choose among multiple augmented-generation techniques or select between different external memory stores. At the collective level, the activities of several LLMs must be coordinated in pursuit of some larger information-process-

ing objective. While these challenges are important, they are not unique to the development of LLM-based personal memory systems; they are, in fact, a standard feature of contemporary efforts to implement multi-agent workflows. For individual LLMs, current approaches to tool and memory-store selection include in-context learning (e.g., Yao et al. 2023), function calling,⁷ and tool-learning techniques (e.g., Schick et al. 2023). In-context learning is the most widely used approach. It involves giving the model structured descriptions of available tools and the conditions under which they should be invoked. Attempts to address the second challenge—the challenge of coordinating multi-LLM activity—typically rely on agentic AI frameworks such as LangChain and AutoGen. These frameworks provide mechanisms for structuring inter-agent communication, delegating subtasks, and managing tool invocations, often through forms of “conversational programming” that treat natural language as a generic interface for agent control (see Wu et al. 2024). As these orchestration frameworks continue to mature, the coordination demands posed by LLM-based personal memory systems are likely to be absorbed into increasingly standardized design practices. Nevertheless, understanding how these orchestration techniques behave in the context of personal memory applications remains an important focus area for future research.

3 Memory Processes

Courtesy of their interpretive, generative, and dialogic capabilities, LLMs can participate in multiple memory processes. In this section, we explore the role that LLMs can play in four key processes, namely, encoding (the process of acquiring information), elaboration (the process of enriching or elaborating on acquired information), retrieval (the process of recalling information), and joint reminiscing (the process of collaboratively re-visiting—and perhaps reconstructing—the past). Our aim here is not to outline the architecture or functionality of a particular memory system; rather, we focus on the contributions that LLMs can make to different memory processes. For the most part, we keep the discussion grounded in the current state-of-the-art, linking the proposed capabilities to the functionality of existing systems.

3.1 Encoding

Encoding refers to the initial capture of information concerning an individual’s activities and lived experience. This process is, of course, supported by conventional lifelogging technologies, such as fitness trackers, smartwatches, body-worn cameras, medical sensors, and so on.⁸ But while lifelogging technologies can capture continuous streams of physiological and behavioral data, LLMs are ideally placed to interpret and annotate this data. An image received from a wearable camera, for example, could be annotated with a linguistic description: “A peaceful solo canoeing trip on a

⁷ See <https://platform.openai.com/docs/guides/function-calling>.

⁸ Some wearable devices provide access to LLMs. Examples include the Rabbit R1 (<https://www.rabbit.tech/>) and Ray-Ban Meta Glasses (<https://www.ray-ban.com/uk/ray-ban-meta-ai-glasses>).

misty morning—light fog on the water and ducks in the distance.” Such annotations also provide opportunities for inference. Assuming the aforementioned image is from a body-worn camera, an LLM can infer that the user is engaged in a canoeing activity (recall the earlier discussion of multimodal LLMs). This capacity for inference is evident in systems such as OS-1, a wearable assistant that integrates smart eyewear with cloud-based multimodal processing (Xu et al. 2024). OS-1 employs a multimodal LLM (LLaVA) to generate scene descriptions from visual input and a speech recognition model (Whisper) to transcribe spoken utterances. The resulting text streams are then used to make inferences about the user’s current activity and location.

Multimodal LLMs thus provide an important opportunity to augment the operation of lifelogging devices, helping to transform low-level sensorimotor data into semantically rich, temporally structured representations of lived experience. Perhaps more importantly, however, LLMs provide an opportunity to move towards a more active and collaborative approach to encoding. To help us see this, consider that some of the information obtained from lifelogging devices may be incomplete or ambiguous. Given an elevated heart rate, for example, it may be hard to determine whether an individual is experiencing stress, anxiety, or excitement. Similarly, an individual’s internal thoughts, motivations, and intentions often leave no trace in the incoming data stream.

To address these problems, an LLM can leverage its dialogic capabilities to actively solicit information from the user. Given the ambiguity surrounding an elevated heart rate, for example, an LLM might pose a question to the user: “How are you feeling right now?”⁹ Similarly, when gaps in the incoming data stream are detected, an LLM can attempt to fill this gap by issuing a request to the user. If, for example, a user is talking about a particular object (“I bought this at the market today”), an LLM can ask the user to take a picture of the object. In such cases, the LLM plays an active role in shaping the encoding process, using targeted prompts to solicit information in a manner that optimizes the prospects for future mnemonic support.

The general idea here is the same as that seen in the earlier cases of RAG, KAG, SAG, and so on (see Sect. 2.3). Just as an LLM can interact with informational and technological resources by generating certain token sequences (e.g., queries, search terms, and programmatic instructions), so it can interact with a human user by capitalizing on its facility with natural language—its ability to generate linguistic expressions that trigger some sort of response from a human individual. The main difference here relates to the nature of the resource that is the target of the LLM’s interaction (i.e., a human individual as opposed to a retrieval system, search engine, or code interpreter). In general, however, we can understand the active shaping of encoding processes in more or less the same way as we understand other types of augmented generation: Just as a well-formed query can surface relevant information from an external datastore, a well-formed question can surface otherwise inaccessible information from the user. In both cases, the goal (for the LLM) is to generate

⁹ Note that there is an important difference between this sort of user-directed querying and the kinds of AI-based emotion inference that have been critiqued in the recent literature (e.g., Goffin & Archer *in press*). Rather than inferring or categorizing a user’s emotional state, the LLM prompts the user to articulate their own feelings, thereby supporting affective self-elaboration without imposing external emotional interpretations.

content (e.g., a question, instruction, or retrieval cue) that optimizes the informational response from an entity that lies external to itself.

The upshot is a novel form of augmented generation that builds on those discussed in the existing literature (see Sect. 2.3). Call this *Human-Augmented Generation (HAG)*. While existing forms of augmented generation focus on the informational and technological aspects of an LLM's ecology (i.e., the resources of the online environment), HAG focuses on the interactions between an LLM and a human individual. Such interactions are, we suggest, likely to be a key feature of the encoding process. The goal of the encoding process is to record information about the lived experience of a human individual, but not all aspects of this lived experience can be distilled from conventional recording devices. In some cases, the LLM needs to 'prompt' the user for information about their cognitive and affective response to situations. Arguably the best way to solicit or (to press the RAG analogy) 'retrieve' this information is to engage the user in conversation.¹⁰ In some ways, such forms of interaction are little different from those in which one LLM communicates with another LLM as part of some larger problem-solving effort (e.g., Schick et al. 2023). In both cases, natural language is being used as a sort of 'programming language' to influence the behavior of another entity.

Ultimately, the goals of encoding for LLM-based memory systems converge with those of conventional lifelogging. That is to say, the goal of encoding is to preserve information in a manner that supports later recall. But the advent of LLMs yields a potentially richer (and more complicated) vision of the encoding process. Multi-modal LLMs provide opportunities for the interpretation and annotation of personal data, while the notion of HAG opens the door to a more interactive and collaborative vision of the encoding process, one in which both LLMs and human users work together to create a richer record of lived experience.

3.2 Elaboration

Encoding results in the storage of information within one or more personal data repositories that function as the external memory for one or more LLMs. Elaboration refers to the processes that enrich and restructure this information.¹¹ This may involve the addition of semantic or contextual information to existing memory records, the removal of redundant content, and the consolidation of newly-acquired information with existing memory records. Some insight into the nature of elaborative processes

¹⁰ Clearly, the timing and frequency of such user-directed engagements must be handled with care, since poorly timed prompts could be intrusive or inappropriate in certain contexts. In practice, these interaction patterns are specified as part of the design of agentic workflows. The AutoGen framework, for example, provides a user-agent component that allows developers to specify the conditions under which an LLM will attempt to solicit human input (Wu et al. 2024).

¹¹ The distinction between encoding and elaboration is somewhat vague, for certain forms of enrichment may occur as part of the encoding process. In the present context, elaboration refers to processes that operate over previously encoded information; e.g., the data contained in one or more personal data stores. The need for elaboration arises as a result of the computational demands of processing high-velocity data streams—such as those associated with physiological sensors (e.g., heart rate data). In addition, some elaborative processes—for example, the generation of textual summaries—are only feasible once a sufficient quantity of data has been collected.

is provided by existing (LLM-based) memory systems. OmniQuery, for example, strives to enhance user memories by establishing semantic and temporal associations between data records (Li et al. 2025). MemoryBank, by contrast, relies on the use of summarization techniques to transform lengthy conversational dialogues into concise memory records (Zhong et al. 2024). Other systems, such as OS-1, exemplify the effort to cluster data into meaningful units, typically by dividing the larger data record into temporal periods marking the occurrence of particular events (e.g., one's attendance at a conference) (Xu et al. 2024).¹²

Elaboration also provides an important opportunity for the creation of narrative artifacts—informational structures that weave personal data into coherent representations of the past.¹³ This is especially relevant given the intimate links between narrative and mnemonic processes in contemporary memory research. In particular, research in psychology and philosophy suggests that autobiographical memory is not merely a store of discrete episodes; it is the basis for constructing an ongoing life story that situates the self within an evolving temporal arc (McAdams 2001; McAdams and McLean 2013). Through processes of selection, organization, and interpretation, individuals craft narratives that explain how they came to be the person they are and where their lives might be heading. This autobiographical narrative is widely regarded as a cornerstone of *narrative identity*—the internalized and evolving story that provides a person with a sense of unity, purpose, and temporal coherence (Schechtman 2005; McAdams 2001).

Narrational capabilities thus represent an important focus area for research into personal memory systems, and it is an area where LLMs are poised to outperform their technological predecessors. Courtesy of their language generation capabilities, LLMs can draw on the data stored in personal data repositories, yielding narratives that are not just factual summaries, but meaningful, temporally structured personal stories that resonate with human users (see Yunusov et al. 2024). In this sense, LLMs are poised to play an increasingly important role in what has been dubbed *narrative niche construction* (Heersmink 2020)—the process by which a variety of mnemonic resources (including narrative artifacts) are created and pressed into cognitive service. This is important when it comes to understanding the distinctive contribution of LLMs to memory-related research. Whereas lifelogging technologies provide us with a vision of data being recorded for a mnemonic purpose, LLMs provide us with a vision in which disparate bodies of data are progressively assimilated into an evolving life story. It is in precisely this sense, we suggest, that LLMs mark a new era in the history of memory technology—one that marks the shift from the mere recording or 'logging' of personal data (i.e., lifelogging) to a process of *life narration*.

¹² Event segmentation is an active area of research for those developing LLM-based memory systems. A recent approach is EM-LLM (Fountas et al. 2025), which applies predictive processing principles to segment token sequences at points of high 'surprise'. The resulting 'event chunks' function as structured episodic units, illustrating how predictive discontinuities can ground artificial episodic memory in ways broadly consistent with human event cognition.

¹³ Some insight into this capability is provided by contemporary AI meeting assistants such as Otter.ai and Fireflies.ai. While these systems tend to produce factual rather than deeply interpretive accounts, they illustrate how existing LLM-based tools can begin to organize complex, temporally extended interactions into coherent narrative-like structures.

It is no doubt natural to assume that the narratives generated by LLMs will be of the textual variety. As noted in Sect. 2.1, however, the advent of multimodal LLMs yields additional possibilities. Using image generation systems, for example, it is possible for life events to be depicted in a visual form, perhaps as a series of images rendered in a comic-book style format. Over time, these synthetic digital artifacts may come to acquire the status of an evocative object—an object that is poised to prompt the recollection of the past in a manner similar to more familiar examples such as photographs, a child’s drawing, or a broken watch (see Heersmink 2022a).

Narratives vary not only in form—such as text or image—but also in temporal scope, ranging from brief, singular events to extended periods of time. Conway and Pleydell-Pearce (2000) propose that autobiographical memory is organized in a hierarchical manner, comprising three levels: lifetime periods (e.g., the years you lived in London), general events (e.g., recurring experiences like Friday dinners at a favorite restaurant), and specific episodes (e.g., a recent trip to Italy for a conference). In principle, LLMs could generate narratives that mirror these hierarchical levels, presenting users with a coherent sense of personal history while allowing for focused retrieval at different scales of granularity. At the time of writing, however, it is unclear how well LLMs perform at generating longer-term or more abstract narratives, such as those associated with lifetime periods and general events. For the most part, extant systems focus on more concrete and temporally-constrained periods, such as those associated with specific events (e.g., Li et al. 2025). In this sense, the hierarchical scheme proposed by Conway and Pleydell-Pearce (2000) may prove useful when it comes to informing the development of future LLM-based memory systems.

3.3 Retrieval

Retrieval is a core feature of memory technologies. If LLMs are to function as personal memory systems, they should not just support the encoding and storage of the past, they should also support the recollection of the past. One way to understand this capability is via the notion of just-in-time information retrieval, where information is retrieved (and presented to a user) in accord with the present context (Rhodes and Maes 2000). Existing LLM-based memory systems, such as Memoro, rely on a mixed initiative retrieval scheme, where information can be retrieved in response to explicit user instructions (e.g., “What was the name of that restaurant we visited last week?”) or information about the present context (Zulfikar et al. 2024).

LLMs can also participate in more interactive and collaborative forms of remembering. One example is cross-cueing, which has been observed in human dyads (Harris et al. 2014). Systems like Tip-pong exemplify this process by engaging users in dialogic exchanges during tip-of-the-tongue states (Ha et al. 2023). When a user struggles to retrieve a specific memory item, the system initiates a conversational back-and-forth—a “ping pong” of semantic prompts and clarifications—until the elusive memory is eventually surfaced. In this case, the LLM does not merely retrieve previously encoded information; rather, it participates in an active cue-elaboration process that combines user-provided cues with its own recollections.

3.4 Joint Reminiscing

Joint reminiscing refers to the process by which earlier memories are revisited (and possibly revised) by LLMs and human users. The resources generated during elaboration are one of the obvious targets for joint reminiscing (see Sect. 3.2). This is particularly true of narrative artifacts, which undoubtedly benefit from some degree of checking and refinement by a human user. As an extension of these processes, users may be involved in the co-production of new narrative artifacts. Examples include life story books (Elfrink et al. 2018) and multimedia biographies (Smith et al. 2009), both of which have been used in clinical settings to help individuals retain a sense of identity in the wake of progressive cognitive decline.

In addition to the collaborative production and editing of narrative artifacts, LLMs are poised to influence memory in a manner similar to that observed in human social contexts. A nice example of this comes in the form of MindTalker—a conversational agent intended for individuals with early-stage dementia (Xyngkou et al. 2024). Built using OpenAI’s GPT-4 model, MindTalker is designed to participate in reminiscence therapy via a user’s photo gallery. In particular, the application engages users in conversations about particular photos by asking open-ended questions such as “Who is in this photo?” or “What do you remember about this day?” This resembles the sort of interventions that are sometimes seen in the case of lifelogging devices, such as SenseCam (see Dubourg et al. 2016). But while these earlier interventions relied on the support of a human caregiver, MindTalker yields a vision in which this socially-supportive role is now occupied by an LLM.

The notion of an LLM occupying a social role points to a broader issue concerning the status of LLMs as memory technologies. Specifically, it highlights how LLMs blur the boundary between technologically- and socially-shaped routes to mnemonic influence. We saw this in the case of Tip-pong, which emulates a process that has traditionally been studied in human–human dyads (i.e., cross-cueing). More generally, by engaging users in conversations about past events or helping them explore the emotional significance of prior experiences, LLMs are poised to recreate the kind of mnemonic scaffolding typically provided by human interlocutors (see Hirst and Echterhoff 2012). Just as conversational interactions with others can help to strengthen, weaken, or modify memories of the past, the conversational capabilities of LLMs may open the door to similar forms of (technological? social?) influence. Addressing these questions is an important area for future research. It expands the scope of existing work by acknowledging the potential of a technological entity to participate in processes of conversational remembering—processes that have typically been confined to the realm of human–human interaction.

4 Component or Companion?

We have discussed some of the ways in which LLMs can contribute to mnemonic processes. But how does this vision of LLMs as a memory technology connect with ongoing philosophical debates about the environmentally-situated nature of human memory processes?

One possibility is that we should regard LLMs as the constituents (or components) of extended cognitive circuits, as per claims regarding extended cognition and the extended mind (Clark and Chalmers 1998; Clark 2010). Relative to this perspective, an LLM would qualify as part of the physical machinery by which certain sorts of mnemonic capacity are realized. Heersmink (2022b), for example, explores how external artifacts—such as diaries, photographs, digital archives, and lifelogging devices—can play a constitutive role in autobiographical memory, functioning as integral parts of remembering processes.

As far as we can tell, there is no reason why LLMs could not qualify as the constituents of extended memory systems. Insofar as we understand memory as a particular sort of capacity—e.g., a capacity to coordinate current thoughts and actions with regard to previously encountered information—then it is at least plausible that LLMs could feature as part of the physical machinery that sustains this capacity. To be a little more precise, we see no reason why LLMs could not form part of the mechanisms that realize memory processes, where a memory process corresponds to the manifestation (or exercise) of a mnemonic capacity that is ascribed to a particular individual (see Smart 2024). In such cases, an LLM would operate as a so-called AI extender (Hernández-Orallo and Vold 2019)—an AI system that qualifies as one of the constituents of an extended cognitive system or an extended mind.

On the other hand, it is far from clear that *every* process in which an LLM is involved ought to be understood as an extended cognitive process. Rather than approach LLMs from an extended perspective, some situations may call for a more embedded approach (Rupert 2004), one that emphasizes the merely causal (but not constitutive) contribution of LLMs to human memory processes. Perhaps, for example, an LLM may qualify as an AI extender when it comes to semantic memory, but fail to qualify as an AI extender when it comes to episodic memory. Rather than regard extended and embedded approaches as competing and mutually exclusive categories, it may be better to adopt a more ecumenical position, one that tailors claims of extended or embedded cognition to the details of particular cases. This is likely to be important given the different ways that LLMs can contribute to human memory processes.

A further complication stems from the status of LLMs as social (or, at any rate, quasi-social) entities (see Clowes *in press*). Consider, for example, the OS-1 system, which is an LLM-based memory system modelled after the fictional character Samantha in the movie *Her* (Xu et al. 2024). In one sense, it is certainly true that OS-1 provides opportunities for cognitive extension, but the main objective of the development effort was not so much to engineer an extended memory system as it was to create an artificial social companion.

There is, of course, no reason to think that social entities could not form part of an extended memory system. It has long been recognized that other human individuals (e.g., a caregiver or intimate companion) could play a constitutive role in cognitive processes, thereby giving rise to what is known as socially-extended cognition (see Clark and Chalmers 1998). The limiting factor, here, is more the extent to which an

extra-organismic resource meets the conditions for cognitive extension¹⁴ than it is the nature of the resource itself. As noted by Clark and Chalmers (1998), appeals to socially-extended cognition are likely to be limited to certain situations, such as those featuring an “unusually interdependent couple.” But these situations need not be so *unusual* once our attention turns to LLMs. Unlike a human companion, an LLM may be more-or-less constantly available and always ready to assist with a cognitive or epistemic task. In this sense, then, LLMs (*qua* social companions) may be much better suited to support socially-extended cognition than their human counterparts. At the same time, of course, an LLM is, at root, a technological entity. Accordingly, LLMs may yield a hybrid form of cognitive extension: one that papers over (or at least blurs) the otherwise crisp conceptual distinction between social and technological forms of mnemonic (and, more generally, cognitive) extension.

5 A ‘Good’ Memory?

A natural concern about the use of LLMs as personal memory technologies centers on their tendency to generate factually incorrect responses—a phenomenon widely referred to as hallucination. Insofar as personal memory systems aim to support the reliable recall of past events, one might reasonably worry that such errors undermine the suitability of LLMs for personal memory applications. One way to mitigate hallucinations is to employ the augmented-generation techniques described in Sect. 2. But while these techniques are effective at reducing errors, none offers a guaranteed solution. In the case of RAG, for example, errors may arise either because incorrect information is retrieved from the external store (retrieval failure) or because information is not properly assimilated into the model’s output (generation failure). Zhang and Zhang (2025) provide a useful overview of these failure modes together with a survey of recent efforts to improve the accuracy and fidelity of RAG-based systems.

While there has been progress towards mitigating hallucinations, it remains unclear whether hallucinations can ever be fully eliminated. The problem is that LLMs are, at root, generative systems: they do not merely retrieve information from internal or external memory but use such information to generate new representations (token sequences) via an inferential process. This naturally invites comparisons with constructivist (or generative) approaches to human memory (Michaelian 2011). Constructivist theories hold that remembering involves the active reconstruction of past events from fragmentary traces, background schemas, and current goals. Because this process relies on inference rather than literal replay, errors and distortions are a recurring—and perhaps unavoidable—aspect of human memory. As Bernstein and Loftus (2009, p. 373) note, “In essence, all memory is false to some degree. Memory is inherently a reconstructive process, whereby we piece together the past to form a coherent narrative that becomes our autobiography.”

¹⁴ The conditions include, according to Clark and Chalmers (1998), the availability of an external resource, the accessibility of the information contained in the resource, and the extent to which the information retrieved from the resource is automatically endorsed.

A propensity for error and distortion thus appears to be a common feature of both LLM architectures and human bio-memory systems.¹⁵ But while LLM hallucinations are widely regarded as flaws or defects—and therefore as problems to be remedied—the treatment of human memory errors is far more nuanced. This is especially true in the case of autobiographical memory, where an expanded appreciation of the cognitive, affective, and social functions served by remembering has prompted a more positive view of mnemonic errors. As Bluck (2003, p. 113) notes, “levels and types of accuracy need not always be regarded as memory ‘failures’ but are sometimes integral to a self-memory system that serves a variety of meaningful ends of human activity.” Conway et al. (2004) make a similar point in their discussion of the self-memory system, arguing that autobiographical remembering is governed by a dynamic tension between adaptive correspondence (the accurate retention of previously encountered information) and self-coherence (the maintenance of a meaningful and temporally integrated sense of self). In many cases, they suggest, the system privileges coherence: thematic elaborations, reinterpretations, and selective omissions that help sustain a stable and intelligible personal identity. Habermas and Bluck (2000) likewise emphasize that narrative identity emerges not from literal fidelity to experience but from the selective integration and interpretive reconstruction of autobiographical memories into a coherent life story. From this perspective, a propensity for error and distortion is not a defect in a system designed for accurate recollection, but a naturally recurring—and often productive—feature of a system geared toward supporting psychological continuity and future-oriented reasoning.

All this, we suggest, complicates the normative interpretation of LLM hallucinations. If human autobiographical memory routinely sacrifices accuracy in favor of coherence, continuity, and psychological well-being, then it is at least plausible that some LLM hallucinations might play a similar role in the context of a personal memory system. Support for this idea stems from work by Sui et al. (2024). They suggest that a propensity for hallucination “mirror[s] a human propensity to utilize increased narrativity as a cognitive resource for sense-making and communication” (p. 14274). Hallucinations, Sui et al. (2024) suggest, are the result of LLMs striving to generate plausible and coherent outputs, as opposed to strictly accurate outputs. This emphasis on coherence at the expense of accuracy is clearly problematic when the goal is to accurately recall the past, but it is not clear that every application of an LLM-based memory system ought to be designed around this goal.

Our aim here is not to underplay the importance of hallucinations in a personal memory context; it is simply to note that we should not judge hallucinations to be problematic in the absence of a clearer understanding of the role that error and distortion play in normal human memory. If the sorts of errors and distortions we observe in normal human bio-memory are understood to play a productive role in the psychological and social lives of individuals, then it is, at best, unclear that technical efforts should be geared to the elimination of those errors and distortions. At the very least, we may need to adopt a more nuanced approach to hallucinations—one that recognizes the hazards they pose for perfect recall while also acknowledging their potential value in the co-creation of a coherent life story.

¹⁵ An additional question—one we do not pursue here—is whether the errors produced by LLMs differ in their structure or impact from those found in human memory, and whether such differences bear on their normative evaluation.

Evidently, there is a need for further research in this area. Such research, however, should not be limited to the standard LLM benchmarks, many of which focus on accuracy and truth. Arguably, what is required are benchmarks specifically tailored to the realm of personal memory—benchmarks that evaluate LLMs in terms of their capacity to optimize the cognitive, affective, and social well-being of their users. Issues of accuracy and truth are clearly important here, but they are not the only concern. Ultimately, what matters is not whether LLMs reproduce the past with perfect precision, but whether they help users navigate and narrate their lives in ways that support cognitive, emotional, and social flourishing. This, we suggest, is the standard by which a genuinely ‘good’ memory system ought to be judged.

6 Conclusion

LLMs fascinate because of their peculiar facility with language. Like us, they can adapt their responses to linguistic instructions and sustain temporally extended conversational exchanges. Such forms of linguistic competence open the door to novel modes of human–machine interaction, enabling us to engage with LLMs in ways that often resemble our interactions with other people. These capabilities are particularly significant in the context of memory applications. As demonstrated by a number of recent systems, LLMs can function as personal memory technologies, retrieving information in response to conversational or contextual cues to yield various sorts of mnemonic support (Xu et al. 2024; Zhong et al. 2024; Zulfikar et al. 2024).

In the present paper, we explored some additional ways in which a facility with language could be used to support the development of personal memory systems. Firstly, we discussed how language opens the door to a more interactive, participatory approach to the encoding of information. Unlike conventional lifelogging technologies, which passively capture data, LLMs are able to shape the structure of the encoding process by (e.g.) requesting additional or clarifying information from a human user. The upshot is a novel form of augmented generation—what we dubbed human-augmented generation. This extends the types of augmented generation that have been discussed in the wider (technical) literature.

We also explored the role of LLMs in generating narrative artifacts. This, we suggested, marks a shift from the notion of lifelogging to life narration. Whereas earlier technologies focused on the ‘logging’ or recording of personal data, LLMs provide opportunities to generate narrative artifacts across a variety of temporal scales. In this respect, LLMs are poised to play an important role in the process of narrative niche construction (see Heersmink 2020).

Finally, we examined the status of LLMs as technological and/or social entities. While LLMs are, at root, technological entities, their proficiency with language enables them to function in the manner of a human social companion. This potentially blurs the distinction between socially-extended and technologically-extended forms of memory. The practical significance of this should be clear: by recapitulating the dynamics of socially-distributed remembering, LLMs yield opportunities to shape human memory in ways that resemble those observed in long-standing human relationships. In addition to the practical considerations, however, LLMs raise inter-

esting questions about the nature of LLM-based memory systems. Are such systems best understood as a technological system that helps us remember the past, or are they best understood as something more akin to a social companion—a new kind of intimate ‘other’ that is poised to provide a variety of types of cognitive, affective, and social support? And who is that intimate ‘other’, exactly? Is the LLM distinct from you, or is it more a part of you—part of the material fabric that makes you the sort of person you take yourself to be? Perhaps, as Clark (2025, p. 3) suggests, such entities will come to be viewed as a ‘borderline you’—an entity that “learn[s] from your projects and choices, and play[s] a role in sculpting your future projects and choices.” In this sense, an LLM is not simply an entity that helps you navigate the past; it is also the co-creator of a shared autobiography. A co-narrator, perhaps, of the story of you.

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